TEXORA · LECTURE SHEET

Jacquard Weaving

The jacquard is the loom's pattern memory: instead of shafts moving ends in groups, every warp end — or every small group — is lifted by its own hook, under the command of punched cards or electronic data. This sheet explains the mechanism end to end, then calculates what a jacquard can and cannot design: hooks against repeats, cards against picks, cost against pattern.

FM-4101 · ADVANCED WEAVING TECHNOLOGY · TOPIC 1 · SPECIALIZATION — FABRIC MANUFACTURING

1. Principle — selection end by end

Tappet and dobby shedding (TE-2103) move warp in shaft groups: every end on a shaft rises together, so the pattern repeat is bounded by the number of shafts. The jacquard removes the shaft: each end is tied by its own **harness cord** to the lifting mechanism, and a pattern device decides, pick by pick, which cords rise. The repeat is then bounded only by the machine's **hook capacity** — hundreds to thousands of independently controlled ends — which is why figured fabrics (damasks, brocades, labels, towels with woven names, upholstery) are jacquard fabrics.

2. The mechanism

PART	FUNCTION
Hooks	Vertical wires, one per controlled end (or per end-group in the tie-up); a hook caught by the rising griffe lifts its harness.
Needles	Horizontal wires, one per hook, spring-loaded; the pattern device pushes selected needles, deflecting their hooks out of the griffe's path.
Griffe (knife block)	Rises and falls once per pick, lifting every hook that has not been deflected.
Cylinder & cards	In the mechanical jacquard, a chain of punched cards, one card per pick, pressed against the needles: a hole lets a needle pass (hook stays, end lifts); a blank pushes the needle (hook deflected, end stays down).
Harness	The set of cords (neck cords, harness cords through the comber board) transmitting each hook's motion to its heddle and warp end.

Lingoes	Small weights on each heddle that pull the end back down and keep the harness taut — the jacquard's return mechanism, one per end.
Electronic jacquard	Solenoid/module selection replaces cards: pattern held as data, changed in minutes, no card cutting, storage or lacing — the modern standard on new installations.

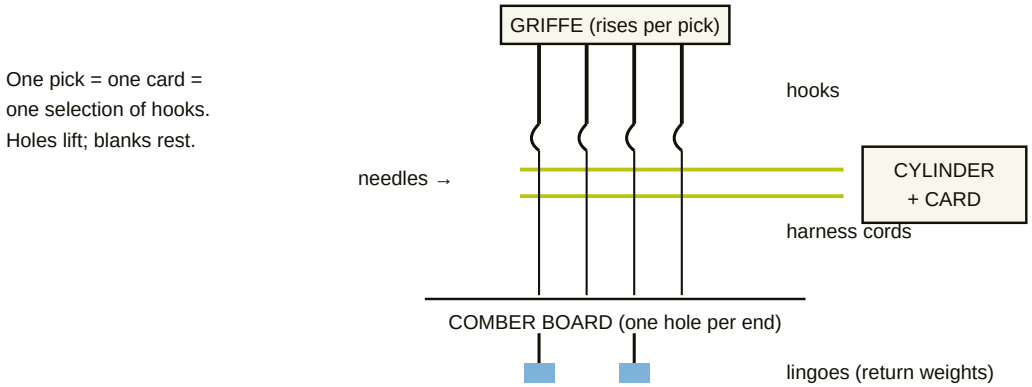


Figure 1 — Jacquard selection in section: card and needles decide, griffe lifts, harness transmits, lingoes return.

3. Capacity calculations — hooks against the design

Ends controlled = hooks × harness repeats across the width

WORKED EXAMPLE — SIZING A DESIGN TO THE MACHINE

A fabric has **5,376 total ends**. The design repeats 4 times across the width, so each repeat carries 5,376 ÷ 4 = **1,344 ends**.

A **1,344-hook** jacquard therefore controls the cloth exactly: each hook governs one end in each repeat through the tie-up.

Had the design needed 1,800 distinct ends per repeat, no tie-up could save it on this machine — the pattern would have to be re-drawn to a smaller repeat, or the machine changed. Pattern width in jacquard weaving is a hook-count statement, and it is made before a single card is cut.

4. Cards — the pattern's length and price

WORKED EXAMPLE — THE CARD CHAIN

Design repeat: 1,800 picks → **1,800 cards** in the chain (one per pick).

At a card pitch of 52 mm the chain is 1,800 × 0.052 = **93.6 m** of laced cards — a physical object that must be cut, laced, stored and eventually disposed of.

Card cutting at Tk 2.5 per card: 1,800 × 2.5 = **Tk 4,500 per design**, before the designer's and

painter's time.

Against this, the electronic jacquard's pattern change is a file transfer. The Tk 4,500 is not the real argument — the 93.6 m chain, its storage, its wear and its changeover time are — but the calculation is how the argument starts.

5. Production

Jacquard looms run slower than plain looms — the selection mechanism and the figured structure both take time — and are costed by the same Sheet 4 arithmetic. At 190 ppm, 72 PPI, 88% efficiency: production = 190 × 60 × 0.88 ÷ 72 ÷ 39.37 × 8 = **28.3 m per shift**. The jacquard premium is never in metres; it is in what the metres can carry — a woven figure no dobby can make at any speed.

6. Mechanical versus electronic — the working differences

ASPECT	MECHANICAL (CARD)	ELECTRONIC
Pattern storage	Physical card chain per design	File; unlimited library
Pattern change	Chain swapped and checked — hours	Data loaded — minutes
Repeat length	Limited by chain practicality	Effectively unlimited picks
Wear parts	Cards, cylinder, needles	Modules/solenoids
Design correction	Re-cut cards	Edit file

Glossary

Hook — vertical lifting wire of the jacquard. **Griffe** — the rising knife block that lifts undeflected hooks. **Tie-up** — the harness arrangement mapping hooks to ends, including repeats across the width. **Comber board** — perforated board guiding harness cords in warp order. **Lingo** — per-end return weight. **Card chain** — the laced sequence of punched cards, one per pick.

Key points

• Jacquard = per-end (or per-group) selection: pattern capacity is hook capacity (worked: 5,376 ends in 4 repeats = 1,344 hooks).

• Selection logic: hole → needle passes → hook lifts; blank → needle pushed → hook deflected → end rests.

• Pattern length is physical on a card machine: 1,800 picks = 1,800 cards = 93.6 m = Tk 4,500 of

cutting.

• Costed like any loom (28.3 m/shift in the example) — the premium is design freedom, not speed.

Practice

1. A 2,688-hook jacquard is tied 6 repeats across the width. How many warp ends can it control, and how many ends are in each design repeat?
2. A design of 2,400 picks is cut on cards at Tk 2.5. Find the cutting cost and chain length at 52 mm pitch.
3. Why can a dobby not weave the 1,344-end repeat of Section 3 at any number of shafts a real dobby offers?

Answer key

1. Ends controlled = 2,688 × 6 = **16,128 ends**; each repeat = **2,688 ends** — the hook count is always the repeat's end count; the tie-up only multiplies it across the width.
2. Cost = 2,400 × 2.5 = **Tk 6,000**; chain = 2,400 × 0.052 = **124.8 m**.
3. A dobby controls shafts — at most a few dozen — and every end on a shaft moves together, so its distinct end-groups number in the dozens, not the thousands. A 1,344-end repeat needs 1,344 independent decisions per pick; that is the definition of jacquard territory.
